BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SARGODHA

INTERMEDIATE (11TH CLASS) EXAMINATION, 2024

MATHEMATICS

Past Paper 2024 • Paper Code: 2191 / 2198

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Q1. Q. 1

Q1. The modulus of Complex number 4 + 5i is

(A) (B)

(C) (D)

Q2. Multiplicative inverse of (2 , 0) is

(A) (B)

(C) (D)

Q3. If A ⊆ B, then A ∩ B equals

(A) (B)

(C) (D)

Q4. Disjunction of two Logical statements p and q is

(A) (B)

(C) (D)

Q5. The solution of linear equation ax = b where a , b ∈ G is

(A) (B)

(C) (D)

Q6. If A = [[1, 2, −3],[0, 2, 0],[−2, −2, 1]], then A_{23} will be

(A) (B)

(C) (D)

Q7. For square matrix A, if $A^t = A$, then A is called

(A) (B)

(C) (D)

Q8. The product of four fourth root of unity is

(A) (B)

(C) (D)

Q9. If α and β are roots of $7x^2 - x - 2 = 0$, then α + β will be

(A) (B)

(C) (D)

Q10. Rational fraction $(x^2 + 2x + 3)/Q(x)$ will be improper fraction if degree of Q(x) is

(A) (B)

(C) (D)

Q11. If in an A.P. $a_1 = 11$, $a_n = 68$, d = 3, then n will be equal to

(A) (B)

(C) (D)

Q12. If 3,9,27,... are in G.P. then r =

(A) (B)

(C) (D)

Q13. The probability of non-occurrence of event E is

(A) (B)

(C) (D)

Q14. The expansion $(1 - 3x)^{1/2}$ will be valid if

(A) (B)

(C) (D)

Q15. If $\cot\theta = 5/2$; $0 < \theta < \pi/2$, then $\text{cosec}^2\theta$ is

(A) (B)

(C) (D)

Q16. $\sin(\theta + 270°) =$

(A) (B)

(C) (D)

Q17. Period of sin x/3 is

(A) (B)

(C) (D)

Q18. $4\Delta/abc$ =

(A) (B)

(C) (D)

Q19. cos (2 $\sin^{-1} x$) will be equal to:

(A) (B)

(C) (D)

Q20. Reference angle always lies in quadrant

(A) (B)

(C) (D)

Q. Section ---------- I

Q2. (Marks 8 × 2 = 16) 2. Answer briefly any Eight parts from the followings:-

(i) Prove that a/b + c/d = (ad + bc)/bd

(ii) Find the multiplicative inverse of (-4 , 7)

(iii) Factorize $9a^2 + 16b^2$

(iv) Prove that product of any two conjugate complex numbers is a real number.

(v) Show that $A - B \subseteq A \cap B'$

(vi) Let (G , •) be a group and a , b ∈ G , then prove that $(a \bullet b)^{-1} = b^{-1} \bullet a^{-1}$

(vii) If A = [[1, −2, 3],[−2, 3, 1],[4, −3, 2]], then find A_{12} and A_{22}

(viii) Given A and B are two non singular matrices, show that $(AB)^{-1} = B^{-1} A^{-1}$

(ix) If A = [[i, 1+i],[1, −i]], then find $A - (\bar{A})'$

(x) Find the fourth roots of unity.

(xi) When $x^3 + 2x^2 + kx + 4$ is divided by x − 2 , then remainder is 14. Find value of k

(xii) Show that the roots of equation $x^2 - 2(m + 1/m)x + 3 = 0$ are real where $m \neq 0$

Q3. (Marks 8 × 2 = 16) 3. Answer briefly any Eight parts from the followings:-

(i) Resolve $(x^2 + 1)/((x - 1)(x + 1))$ into partial fraction

(ii) Define conditional equation.

(iii) Determine whether -19 is term of A.P 17,13,9,...

(iv) Find geometric mean between $-2i$ and $8i$

(v) Sum the infinite geometric series $4 + 2\sqrt{2} + 2 + \sqrt{2} + \ldots$

(vi) Find 12th term of H.P 1/3 , 2/9 , 1/6 ,...

(vii) Evaluate ${}^{10}p_7$

(viii) How many ways can 4 keys be arranged on a circular key ring.

(ix) How many diagonals can be formed by joining vertices of 5 sided figure

(x) Expand $(x - i - 1/x)^3$

(xi) Expand upto four terms $(1 + x)^{-3}$

(xii) Find term involving x^5 in expansion of $(x^2 - 3/2x)^{10}$

Q4. (Marks $9 \times 2 = 18$) 4. Answer briefly any Nine parts from the followings:-

(i) Express 75° in radians.

(ii) Prove that $\sin\theta/(1 + \cos\theta) + \cot\theta = \operatorname{cosec}\theta$

(iii) If α,β,γ are angles of a triangle, then prove that $\cos((\alpha + \beta)/2) = \sin\gamma/2$

(iv) Without using calculator, find the value of tan 105°.

(v) Prove that $(1 - \cos\alpha)/\sin\alpha = \tan\alpha/2$

(vi) Write the domain and range of $y = \cos x$

(vii) Define periodicity.

(viii) Find the period of $3\cos x/5$

(ix) At the top of a cliff 80 m high, the angle of depression of a boat is 12°. How far is the boat from the cliff?

(x) Find area of a triangle ABC in which $a = 18$, $b = 24$, $c = 30$

(xi) Show that $r_2 = s\tan\beta/2$

(xii) Show that $\cos(\sin^{-1} x) = \sqrt{(1 - x^2)}$

(xiii) Solve the equation $1 + \cos x = 0$ for general solution.

Q. Section ---------- II

Note: Attempt any three questions. ($10 \times 3 = 30$)

Q5. 5. (a) Find the inverse of the matrix A = [[2, 5, −1],[3, 4, 2],[1, 2, −2]]

(b) Solve the system of equations $y^2 - 7 = 2xy$ $2x^2 + 3 = xy$

Q6. 6. (a) Resolve $x^4/(1 - x^4)$ into Partial Fractions.

(b) The A.M of two positive integral numbers exceeds their (positive) G.M by 2 and their sum is 20, find the numbers.

Q7. 7. (a) Prove that ${}^{n-1}C_r + {}^{n-1}C_{r-1} = {}^nC_r$

(b) If $y = 1/3 + 1.3/2!(1/3)^2 + 1.3.5/3!(1/3)^3 + \ldots$ then prove that $y^2 + 2y - 2 = 0$

Q8. 8. (a) Reduce $\cos^4 \theta$ to an expression involving only function of multiples of θ, raised to the first power.

(b) Prove that $r_1 + r_2 + r_3 - r = 4R$

Q9. 9 (a) Prove that $\sqrt{((1 - \sin \theta)/(1 + \sin \theta))} = \sec \theta - \tan \theta$, where θ is not an odd multiple of $\pi/2$

(b) Prove that $\sin^{-1} 77/85 - \sin^{-1} 3/5 = \cos^{-1} 15/17$

GROUP – II

Q1. Q. 1

Q1. The transpose of a rectangular matrix is a

(A) (B)

(C) (D)

Q2. $1 - \omega + \omega^2 =$

(A) (B)

(C) (D)

Q3. The quadratic equation with roots $3-\sqrt{3}$, $3+\sqrt{3}$ is

(A) (B)

(C) (D)

Q4. The reflexive property of equality of real numbers is that $\forall$ a $\in$ R

(A) (B)

(C) (D)

Q5. $|Z|^2 =$

(A) (B)

(C) (D)

Q6. $\{x|x \in N , x \leq 10\}$ is the

(A) (B)

(C) (D)

Q7. $p : 4 < 7 , q : 6 > 11$, the disjunction $p \vee q$ is

(A) (B)

(C) (D)

Q8. The identity element of a set X with respect to intersection in P(X). is

(A) (B)

(C) (D)

Q9. If $A = [[x, 1],[1, 1]]$ and $1/|A| = 7$, then x =

(A) (B)

(C) (D)

Q10. $r_1r_2r_3 =$

(A) (B)

(C) (D)

Q11. $2\cos^{-1} A =$

(A) (B)

(C) (D)

Q12. $\cos x = -1/\sqrt{2}$ and $x \in [0, \pi]$ then x =

(A) (B)

(C) (D)

Q13. $(x - 4)^2 = x^2 - 8x + 16$ is

(A) (B)

(C) (D)

Q14. A number A is said to be the arithmatic mean between two numbers a and b if a , A , b is

(A) (B)

(C) (D)

Q15. If a = 3 , r = 2 then nth term of the G.P is

(A) (B)

(C) (D)

Q16. $n(n-1)(n-2)(n-3)...(n-r+1) =$

(A) (B)

(C) (D)

Q17. The sum of the odd coefficients in the expansion $(1 + x)^3$ is

(A) (B)

(C) (D)

Q18. $120° =$ _____ radians

(A) (B)

(C) (D)

Q19. $2 \sin^2(\alpha/2) =$

(A) (B)

(C) (D)

Q20. The range of sin x is

(A) (B)

(C) (D)

Q. Section ---------- I

Q2. (Marks 8 × 2 = 16) 2. Answer briefly any Eight parts from the followings:-

(i) Prove the rule of addition $a/b + c/d = (ad + bc)/bd$

(ii) Separate real and imaginary parts $(2 - 7i)/(4 + 5i)$

(iii) Find the multiplicative inverse of $-3 - 5i$

(iv) For any complex number $z \in C$, prove that $z \cdot \bar{z} = |z|^2$

(v) If $S = \{0 , 1 , 2\}$, then show that S is an abelian group under addition.

(vi) Construct the truth table of the statement $(p \wedge \sim p) \rightarrow q$

(vii) If $B = [[5, -2, 5],[3, -1, 4],[-2, 1, -2]]$, then find B_{21} and B_{23}

(viii) If A is symmetric or skew-symmetric, show that A^2 is symmetric

(ix) Find the matrix X if $X[[5, 2],[-2, 1]] = [[-1, 5],[12, 3]]$

(x) Show that the product of all the three cube roots of unity is unity.

(xi) If α , β are the roots of $x^2 - px - p - c = 0$, prove that $(1 + \alpha)(1 + \beta) = 1 - c$

(xii) Solve the equation $x^4 - 6x^2 + 8 = 0$

Q3. (Marks 8 × 2 = 16) 3. Answer briefly any Eight parts from the followings:-

(i) Define a Rational Fraction with example.

(ii) Resolve into partial Fraction without determining the constants $(3x^2 - 4x - 5)/((x - 2)(x^2 + 7x + 10))$

(iii) If 1/a , 1/b and 1/c are in A.P, show that $b = 2ac/(a + c)$

(iv) If $S_n = n(2n + 1)$, then find the series

(v) A.M between two numbers is 5 and their positive G.M is 4. Find the numbers.

(vi) If 5 is Harmonic Mean between 2 and b. Find b

(vii) Find the value of n , when ${}^{n}P_4 : {}^{n-1}P_3 = 9 : 1$

(viii) A die is rolled, what is the probability that the top shows dot 3 or 4.

(ix) Find the number of the diagonals of a 6 − sided figure.

(x) State the principle of Mathematical induction.

(xi) Prove the formula $2 + 4 + 6 + \ldots 2n = n(n + 1)$

(xii) Find the general term of $(a/2 - 2/a)^6$

Q4. (Marks 9 × 2 = 18) 4. Answer briefly any Nine parts from the followings:-

(i) State fundamental identities.

(ii) Verify that $\sin^2 \pi/6 : \sin^2 \pi/4 : \sin^2 \pi/3 : \sin^2 \pi/2 = 1 : 2 : 3 : 4$

(iii) Prove that $\cos 330° \sin 600° + \cos 120° \sin 150° = -1$

(iv) Show that $\cot(\alpha + \beta) = (\cot \alpha \cot \beta - 1)/(\cot \alpha + \cot \beta)$

(v) Prove that $\sin(\alpha + \beta) - \sin(\alpha - \beta) = 2 \cos \alpha \sin \beta$

(vi) Write down the Domain and Range of secant function.

(vii) Find the period of tan 4x

(viii) Draw the graph of y = sin x from 0 to π

(ix) Define the angles of elevation and depression.

(x) What do you mean by oblique triangle.

(xi) By using law of cosine, find α when a = 7 , b = 3 , c = 5

(xii) Prove that $\sin^{-1} x = \pi/2 - \cos^{-1} x$

(xiii) Solve the trigonometric equation $\cot^2 \theta = 1/3$

Q. Section ---------- II

Note: Attempt any three questions. (10 × 3 = 30)

Q5. 5. (a) Use Crammer's Rule to solve the systems of Linear equations: $3x_1 + x_2 - x_3 = -4$ $x_1 + x_2 - 2x_3 = -4$ $-x_1 + 2x_2 - x_3 = 1$

(b) Find the values of a and b if -2 and 2 are the roots of the polynomial $x^3 - 4x^2 + ax + b$

Q6. 6. (a) Resolve into partial fractions $(x^2 + 2x + 2)/((x^2 + 3)(x + 1)(x - 1))$

(b) How many terms of the series -9 -6 -3 + 0 + ... amount to 66?

Q7. 7. (a) Find values of n and r when ${}^{n-1}C_{r-1} : {}^{n}C_{r} : {}^{n+1}C_{r+1} = 3 : 6 : 11$

(b) If $2y = 1/2^2 + 1.3/2!\cdot 1/2^4 + 1.3.5/3!\cdot 1/2^6 + ...$ then prove that $4y^2 + 4y - 1 = 0$

Q8. 8. (a) Prove that $\sin 10° \cdot \sin 30° \cdot \sin 50° \cdot \sin 70° = 1/16$

(b) Using Law of tangents, solve the ΔABC in which a = 36.21 ; c = 30.14 ; $\beta = 78°10'$

Q9. 9 (a) If $\operatorname{cosec} \theta = (m^2 + 1)/2m$; $m > 0$; $0 < \theta < \pi/2$, then find the values of remaining trigonometric functions.

(b) Prove that $2 \tan^{-1}(1/3) + \tan^{-1}(1/7) = \pi/4$